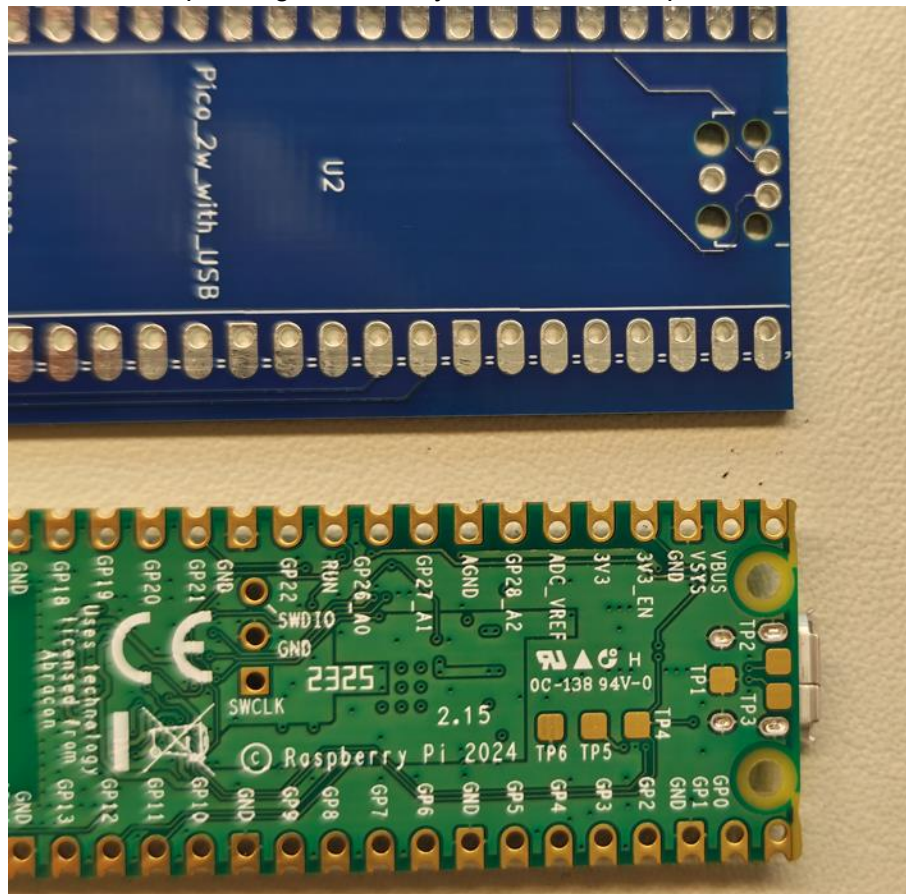
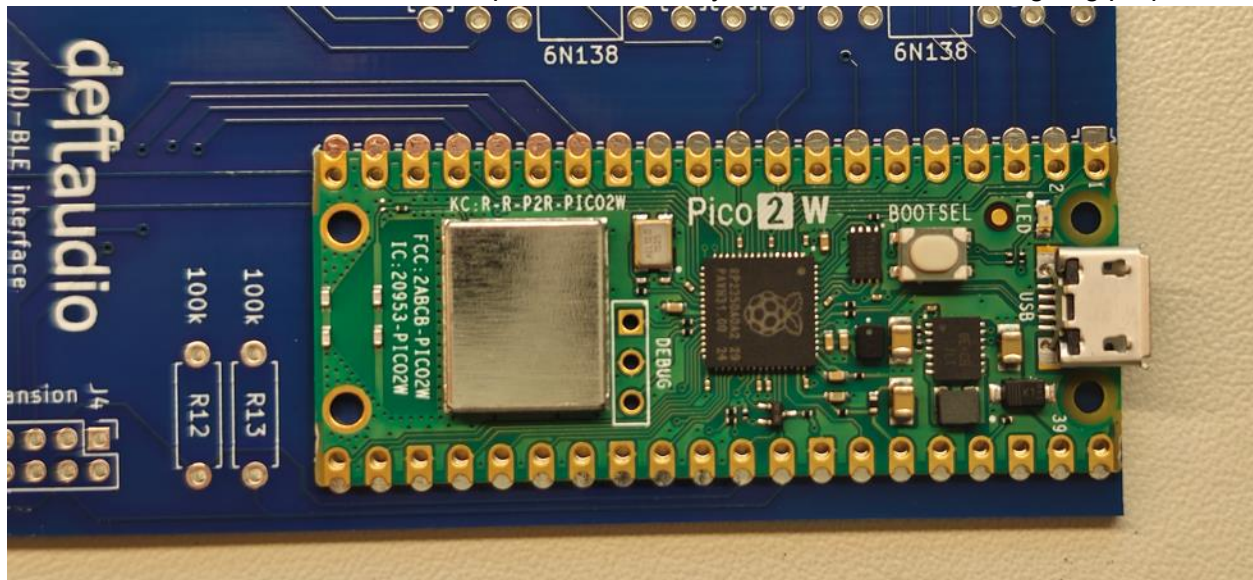


Beginner soldering skills required. That means you have to understand schematics, not just being able to solder two wires together. There are no high expectations, but being able to read schematics and understand the polarity of components are quite important.

Solder Raspberry Pi Pico 2W. This part is a bit tricky, since the module uses USB connections on the back side of the board. Those are pin pads and marked as TP1, TP2 and TP3. On the main PCB there are corresponding holes that you need to feel up with a solder. See a picture:



It's easier to solder the top pins first. Carefully align the holes of Pico and a footprint on the PCB. You don't need to solder it with pin headers, but you can use those for aligning purpose.



Then flip the board and solder TP1-TP3 by filling up holes with enough solder (better to add a drop of flux so it will go all in).

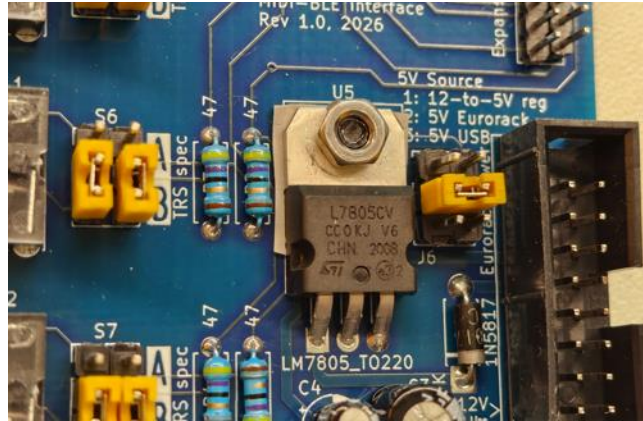


Step 3:

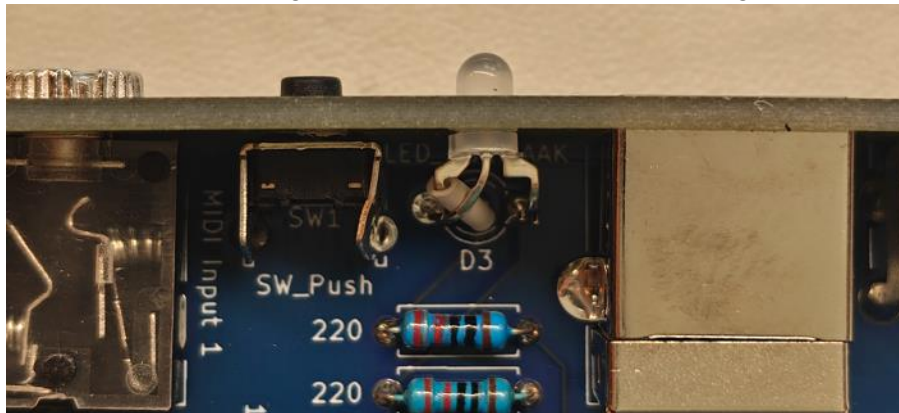
Solder the rest of the parts starting with IC sockets, capacitors, and everything else except LM7805 power regulator and the LED.

Step 4:

Solder LM7805. Add a thermal pad under it and mount with m3 screw and nut.

**Step 5:**

Solder bi-color LED. This piece is tricky too. The frosted LED has a different pinout and requires flip pins. This way the central pin goes to Pin 1. Make sure you isolate leads so they don't make a short (a piece of wire isolation is enough). Also note the LED is not symmetrical one leg is rounded while another has a 90-degree lead which has to be on the right. Refer to the photo:

**Step 6:**

Final step - attach jumpers. Refer to silkscreen for TRS-A/TRS-B position for MIDI ports. J6 sets the supply voltage - from local LM7805 5V regulator (1), from Eurorack 5V rail (2) or from USB input (3).

Software Updates:

Firmware programming is not required as all kits come with pre-programmed Pico 2W. New firmware posted at https://github.com/Deftaudio/Midi-boards/tree/master/Eurorack_MIDI-BLE_2x2/Firmware

To update firmware hold BOOTSEL on Pico board while turning it on and connect to computer via USB cable. Pico will appear as the USB disk, copy UF2 file on it.

Operation Modes

A front-panel button and status LED let you choose how your gear interacts:

Green mode - clean, direct pass-through: USB port 1 talks to TRS 1, and TRS 2 becomes a shared meeting point for USB port 2 and your wireless connection.

Red mode - adds a full merge between your two TRS ports, so hardware gear on either jack sees the combined stream of both - perfect for daisy-chaining a small hardware rig without a separate merge box.

Status LED is blinking while BLE is enabled and the device is discoverable. Once the BLE connection is established, it turns solid.

Status LED also blinks on any MIDI traffic.